

Commons-Governed Analogy Sweep as Public-Interest Discovery Infrastructure

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Bounded AI Mapping, Provenance, and Anti-Enclosure Design

Method and Evidence Boundary

This is a conceptual methods and governance paper, not an empirical evaluation, patent-law article, or claim of autonomous machine discovery. Its defensible contribution is a bounded workflow for making AI-assisted analogy generation more reviewable, provenance-bearing, and commons-oriented. The necessary evidence is correspondingly modest: a clear method, careful non-claims, relevant literature on analogy and knowledge commons, and one illustrative sweep artifact that shows how the method would be recorded.

The paper draws on anonymized source-project materials that define a structural profiling layer, a cross-domain and cross-scale analogy sweep, a conceptual translation layer, value/risk scoring, commons-publication intent, and explicit non-agency and legal-boundary notes. Those materials are used as design evidence, not as proof that the method improves discovery outcomes. User studies, expert scoring panels, legal review, or large-scale output audits would support stronger later claims, but they are not prerequisites for the narrower article proposed here.

Use of Generative AI

OpenAI ChatGPT and OpenAI Codex assisted with drafting, structuring, literature targeting, and document preparation. The author reviewed and revised all generated content and remains responsible for the manuscript's claims, source interpretation, legal-boundary framing, and final submission decisions.

Abstract

Generative AI systems can rapidly expand the implication space around an idea by producing analogies, adjacent applications, domain translations, and speculative mappings. This capacity can widen public reasoning, but it can also intensify enclosure if cross-domain ideation becomes a proprietary service controlled by institutions with greater compute, data access, and legal capacity. This paper proposes commons-governed analogy sweep as a public-interest method for AI-assisted conceptual expansion. Drawing on anonymized source-project materials and related signal-atlas practices, it defines a bounded workflow: profile an originating idea by properties, sweep across domains and scales, translate structural relations into candidate application maps, score candidate outputs for public value and risk, preserve provenance, and publish outputs as commons-oriented artifacts where lawful and appropriate. The method does not treat AI systems as autonomous discoverers, scientific validators, patent-search tools, or legal authorities. Candidate mappings remain invitations to evaluate. Humans retain novelty judgment, validation responsibility, publication choice, and legal review. The paper's contribution is a governance

pattern for analogy-generating systems: expand access to structured reasoning while constraining overclaim, documenting uncertainty, and treating public disclosure as a design goal. It argues that the value of such systems should be assessed by whether they lower barriers to implication-space exploration, preserve human authority, expose blind spots, and produce traceable artifacts that strengthen the knowledge commons rather than private capture.

Keywords: artificial intelligence; analogy; knowledge commons; public interest technology; governance; provenance

1. Introduction

Discovery is not evenly distributed. Institutions with legal teams, research staff, proprietary databases, and large technical systems can explore implication spaces that independent researchers, artists, small teams, and public-interest builders often cannot inspect. A person may have a useful idea but lack the time, language, disciplinary access, or tooling to ask where else the idea might apply. This inequality matters in an era when generative AI can rapidly produce analogies, translations, summaries, and cross-domain maps.

The question is not only whether AI can generate more ideas. It is who benefits from the expansion of possibility space. If machine-assisted analogy work is captured inside proprietary systems, the same capacity that could widen participation may become another enclosure layer. If it is designed for public disclosure, provenance, and human review, it may help more people inspect where their ideas travel.

This paper develops commons-governed analogy sweep as a public-interest method. The anonymized source framework combines structural profiling, cross-domain sweep, conceptual translation, and value/risk scoring. In journal-safe terms, these are best understood as components of a structured reasoning protocol: profile an idea by properties, search across domains and scales, translate local relations into candidate application maps, and score outputs for human assessment.

The risky language in the source project is also important. Terms such as "universal," "engine," "machine," and "humanity owns it" carry strong rhetorical force. A journal paper should not treat them literally. The source framework is not an omniscient search engine, legal oracle, autonomous scientist, patent office, or guarantee of novelty. It is a bounded process for making candidate analogies more visible and more governable.

The central contribution of this paper is therefore a governance pattern. AI-assisted analogy tools should be judged not by whether they "discover" in a strong autonomous sense, but by whether they lower access barriers to structured reasoning, preserve human authority, document candidate mappings, expose uncertainty, and publish outputs in forms that strengthen the knowledge commons.

2. Enclosed Implication Space

Ideas often matter beyond their first domain. A technical mechanism may have implications for education, accessibility, climate modeling, interface design, public administration, or safety. A social practice may suggest a software pattern. A biological metaphor may inspire a workflow method. A local design constraint may reveal a general governance principle. Much of research and invention depends on seeing these movements.

Implication space is expensive to search. Patent landscapes, prior-art databases, academic literatures, standards documents, and field-specific vocabularies create access barriers. Search engines retrieve strings; expert databases presuppose domain knowledge; institutional analysts can privatize promising connections before they become public. Smaller actors may sense a useful structure but fail to document adjacent applications before a better-resourced actor encloses a version of it.

Generative AI changes this terrain. Language models can propose analogies and application domains quickly. Used carelessly, this creates noise, false confidence, and speculative overreach. Used carefully, it can lower the cost of exploratory reasoning. The governance question is where authority sits. Does the system claim to discover, own, validate, and privatize? Or does it help humans map candidate spaces that remain open to scrutiny?

Commons-governed analogy sweep takes the second path. The system reorganizes existing knowledge through structural analogy and property mapping. It may surface candidate mechanisms or analogies for humans to assess, but it does not validate new science by itself. A candidate mapping is an invitation to evaluate, not a conclusion.

3. Source Project and Method

The source framework defines a structured reasoning and commons-publication workflow. Its project materials describe four main components:

1. A structural profiling layer for concepts and systems.
2. A cross-domain, cross-scale analogy sweep.
3. A Conceptual Translation Layer that converts local ideas into broader application maps.
4. A value/risk scoring layer for high-value, lower-risk candidates.

The same source materials also contain a boundary note: the system is not an autonomous discovery agent, legal oracle, or safety guarantee. Its commons language is explicitly framed as project doctrine and license intent, with enforceability and patent effect depending on jurisdiction and facts. This boundary is essential for any publishable version of the work.

The related signal-atlas materials provide a second useful source. They do not constitute the analogy-sweep system itself, but they model the kind of cautious artifact discipline this paper recommends: entries are treated as candidate signals, correspondences, mechanism traces, tooling lenses, background, or exclusions; tool-derived outputs are separated from direct source material; and uncertain links are downgraded rather than promoted.

Taken together, these sources support a method with five stages:

1. Profile the originating idea by properties, relations, constraints, inputs, outputs, and failure modes.
2. Sweep across domains and scales for candidate structural analogies.
3. Translate local features into candidate application maps.
4. Score candidates for public value, feasibility, risk, evidential strength, novelty uncertainty, and governance sensitivity.
5. Publish or archive outputs with provenance, scope, uncertainty labels, and human review state.

4. Analogy Sweep Is Not Similarity Search

Analogy sweep is related to semantic search, but it is not identical to it. Similarity search finds nearby terms, documents, or embeddings. Analogy sweep asks whether a structural relation may travel across contexts. A query about "routing ambiguous material into an overflow queue" might connect to pedagogy, interface triage, medical referral, software error handling, and administrative escalation even when those domains do not share obvious keywords.

The method is also not patent search. Patent search has legal, jurisdictional, claim-construction, enablement, novelty, and non-obviousness standards that an analogy-sweep workflow does not satisfy by itself. A structured analogy artifact may later help a human or lawyer identify relevant prior art, but the system should not present its output as a legal determination.

The method is also not autonomous discovery. It can widen the visible field around an idea. It cannot decide that a mapping is true, safe, novel, lawful, useful, or ethically appropriate. Those judgments remain human and institutional responsibilities.

The paper therefore uses "candidate mapping" as the central output category. A candidate mapping is neither nothing nor proof. It is a documented possibility with provenance, rationale, uncertainty, risk, and review status.

5. Human Authority and Non-Agency

The non-agency boundary is central. A responsible analogy-sweep workflow should separate roles:

1. The human supplies the originating idea, purpose, and constraints.
2. The system profiles properties and proposes candidate domains.
3. The system records uncertainty, risk, and weak evidence.
4. The human evaluates whether a mapping is meaningful.
5. Domain experts, experiments, or legal review handle validation where needed.
6. Outputs are published or archived with provenance, scope, and disclaimers.

This division makes the tool useful without overstating it. Good analogies reveal structure. Bad analogies launder vagueness. A language model can generate impressive lists of cross-domain parallels that lack operational bite. Without human assessment, analogy sweep becomes a rhetoric amplifier.

Human authority does not mean unstructured authority. The human reviewer should also leave traces: acceptance, rejection, uncertainty, domain-expert-needed status, legal-review-needed status, safety-review-needed status, and reasons for downgrading weak analogies. The aim is not merely to generate more ideas, but to produce reviewable knowledge artifacts.

6. Artifact Schema for Public Sweeps

Commons-governed analogy sweep becomes publishable only if its outputs are legible. A useful public artifact should contain enough information for later readers to understand what was claimed, what was not claimed, and how the mapping was produced.

Artifact field	Purpose
Originating idea	States the starting concept or system without inflating it
Property profile	Records functions, constraints, relations, inputs, outputs, and failure modes
Sweep configuration	Names model, prompt version, date, domains, scale range, and exclusions
Candidate mappings	Lists analogies and application maps as provisional outputs
Structural rationale	Explains which relation is claimed to map
Breakpoints	Records where the analogy fails or becomes weak
Value/risk score	Separates public value, feasibility, risk, uncertainty, and governance sensitivity
Provenance	Provides timestamp, version, repository, license, and artifact hash where possible
Review state	Marks unreviewed, author-reviewed, expert-reviewed, rejected, or publication-ready
Legal boundary	States that the artifact is not legal advice and does not guarantee patent effect

This schema does two things. First, it improves epistemic discipline by forcing weak links and uncertainty into the record. Second, it makes publication more useful. A later reader does not merely see a claim that an idea could apply elsewhere; they see the route by which that candidate mapping was produced.

7. Illustrative Sweep Artifact

The following compact artifact is illustrative rather than evidential. It shows the minimum record a commons-governed analogy sweep should leave behind.

Artifact field	Illustrative entry
Originating idea	A host-layer queue that routes uncertain or unsafe generated claims away from immediate publication.
Property profile	Ambiguous input; downstream consequence if mishandled; need for triage, delay, provenance, and human review; failure mode of fluent premature certainty.
Sweep configuration	Conceptual sweep across education feedback, clinical referral, software issue triage, public-record annotation, and scholarly peer review; legal and safety domains flagged as expert-review-required.
Candidate mapping	Treat uncertain generated claims like triage cases: they are not discarded, but routed into a review lane with status, evidence, and escalation criteria.
Structural rationale	The relation that travels is not surface similarity but controlled deferral: a provisional item becomes safer when the system records why it is uncertain, who can act on it, and what evidence would move it forward.
Breakpoints	Clinical, legal, and safety uses require domain authority; the analogy fails if the system presents deferral as validation or uses review queues to hide accountability.
Value/risk score	Public value: medium-high; feasibility: high for documentation systems; risk: medium where delay affects rights or safety; novelty uncertainty: high; governance sensitivity: high.
Provenance	Author-reviewed illustrative artifact prepared for this manuscript; source framework anonymized for review; final repository or archive status to be resolved after review.
Review state	Author-reviewed example; not expert-validated; not legal advice; not a claim of prior-art effect.
Publication posture	Suitable as a manuscript example and possible supplement if anonymized; broader public archive requires final source-release and licensing decisions.

8. Commons Doctrine and Anti-Enclosure

The most distinctive part of the source framework is its commons doctrine. In project language, machine-assisted candidate mappings should default toward public benefit rather than private capture. Its license addendum classifies outputs as integral system artifacts, intends them to inherit reciprocal commons obligations under project doctrine, and frames public disclosure as a way to resist patent enclosure and proprietary lockup to the maximum lawful extent.

A journal article must translate this carefully. The addendum records licensing intent and project doctrine. It is not legal advice. Enforceability, patent effect, prior-art status, ownership, and contractual interaction can depend on jurisdiction, publication facts, authorship questions, claim scope, and applicable law. The paper should not claim that an addendum automatically prevents patents or makes every output legally unenclosable.

Even with that caution, the governance idea is significant. AI-assisted ideation can create enclosure pressure. If a tool rapidly maps applications across domains, the entity controlling the tool may gain an advantage in filing patents, building proprietary implication-map databases, or selling closed discovery-as-a-service. A commons-governed alternative can push in the opposite direction by normalizing public publication, provenance, and reciprocal openness.

This is best understood as infrastructural design rather than legal certainty. The goal is to make commons publication the default social pathway for machine-assisted conceptual expansion.

9. Public Prior Art as Design Practice

Prior art is often discussed after a patent claim must be challenged. Commons-governed analogy sweep suggests a proactive practice: publish structured candidate mappings early enough, clearly enough, and openly enough that later enclosure may become harder where law and facts recognize the disclosure.

The practice is not a magic shield. But it has practical requirements. Public artifacts should have stable timestamps, accessible URLs, clear descriptions, versioning, author/source provenance, license notices, and bounded claims. They should avoid saying "the system discovered X" when the safer claim is "the system generated candidate mappings that a human may evaluate." They should avoid saying "this prevents patents" when the safer claim is "this publication is intended to support commons access and may function as prior-art disclosure depending on facts and law."

The value is partly legal, partly epistemic, and partly political. Legally, public disclosure can matter. Epistemically, structured artifacts make reasoning inspectable. Politically, commons publication contests the idea that machine-assisted exploration should default to private capture.

10. Future Validation Agenda

The current source materials support a conceptual methods and governance paper, not a completed empirical evaluation. That boundary should remain visible. The following dimensions would strengthen later versions if the article claimed measured effectiveness, expert acceptance, or practical legal utility.

Method reproducibility asks whether independent users can reconstruct the property profile, candidate domain sweep, conceptual maps, scores, and exclusions from the same input and configuration.

Mapping quality asks whether domain reviewers judge candidate analogies as structurally relevant, novel, useful, risky, or overextended. Negative cases should be included.

Access impact asks whether non-institutional users identify more relevant domains, literatures, or applications after using the tool than before. Measures might include domain recall, search diversity, expert-rated maps, or confidence calibration.

Prior-art readiness asks whether output artifacts contain stable timestamps, clear descriptions, provenance, publication status, license notices, and legal-boundary statements. Legal experts could assess usefulness while preserving the caveat that no general guarantee is possible.

Risk handling asks whether the system identifies dangerous applications, speculative leaps, privacy risks, dual-use concerns, and domains requiring expert validation. A tool that expands implication space without expanding caution would be irresponsible.

These are desirable future validation routes, not submission prerequisites for the smaller claim advanced here.

11. Relation to Responsible Technology

Commons-governed analogy sweep sits in a broader debate about AI as public good. Many AI systems are described as productivity tools, but their deeper effect may be on who gets to explore possibility space. If only large firms can run broad ideation, analogy mining, and implication mapping at scale, AI may intensify asymmetries in invention. If public tools can expose structured candidate maps for ordinary users, AI may support a more democratic discovery culture.

Responsible technology requires more than warnings. It requires architectures that allocate power differently. Commons-governed analogy sweep proposes several such allocations: human users retain interpretive authority; the system generates candidate maps rather than settled claims; outputs carry provenance and license intent; risk scoring is part of the method; and public publication is treated as a design goal.

These choices make the method relevant to AI ethics even though it is not primarily a model-safety tool. It addresses the social fate of machine-assisted conceptual expansion.

12. Limitations

The first limitation is overgeneration. Cross-domain sweep can produce many weak candidate analogies. The method needs pruning, scoring, and exclusion discipline.

The second limitation is false structural fit. An analogy can share surface properties while failing at the mechanism level. Reviewers should state which relation maps and which parts break.

The third limitation is legal uncertainty. Commons doctrine and license addenda can express intent and impose obligations on covered uses, but they do not automatically settle patent law, ownership, enforceability, or prior-art status across jurisdictions.

The fourth limitation is strategic misuse. A public analogy tool could be used by private actors to identify targets for enclosure. Commons licensing and publication can help but cannot eliminate all extractive use.

The fifth limitation is unequal validation capacity. Generating candidate mappings is easier than testing them. Public-interest discovery infrastructure must avoid dumping unvalidated possibility space onto people who lack the resources to evaluate it.

The sixth limitation is rhetoric. Terms such as universal, machine, enforcement, and humanity owns it can inspire, but they can also overstate. A journal version should preserve them as project vocabulary only when surrounded by careful boundary language.

13. What This Paper Is Not Claiming

This paper does not claim that AI systems autonomously discover new science.

It does not claim that analogy sweep validates hypotheses, grants novelty, or performs patent search.

It does not claim that AGPL licensing or a commons addendum automatically prevents all enclosure or patent claims.

It does not provide legal advice.

It does not claim that all analogies are useful or that expanding implication space is always beneficial.

It does not claim that public disclosure is always appropriate; privacy, safety, consent, and community interests may require withholding or redaction.

14. Conclusion

Commons-governed analogy sweep is a modest but valuable idea. It does not make machines into discoverers. It does not validate science. It does not guarantee legal protection. Its promise lies elsewhere: it helps humans profile ideas by structure, explore adjacent domains, expose blind spots, and publish candidate maps as public artifacts.

In an era when AI can expand implication space quickly, the ownership and publication model matters. If machine-assisted reasoning defaults to private capture, it may deepen existing asymmetries. If it is designed as public-interest infrastructure, it can help more people see where their ideas may travel while keeping validation, responsibility, and benefit in human and public hands.

The strongest claim is not that the machine finds and humanity automatically owns. The stronger publishable claim is that systems can be designed so machine-assisted exploration produces bounded, provenance-bearing, reviewable artifacts whose default orientation is the knowledge commons.

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